

Multi-Task Learning for Bangladeshi Banknote Denomination and Condition Recognition

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Abstract—Reliable banknote recognition under everyday mobile capture conditions is difficult because illumination, background, viewpoint, device characteristics, and physical wear can all vary simultaneously. This study introduces a self-collected Bangladeshi banknote image dataset constructed for two complementary tasks: denomination recognition and note condition classification. The dataset covers seven circulating denominations and three condition labels collected under daylight and low-light conditions, unconstrained backgrounds, and both old and new note categories. Using stratified five-fold cross-validation and training-only offline augmentation, this study evaluated two transfer-learning strategies: independent task-specific networks and a compact MobileNetV3-Large backbone with a shared multi-layer representation and two prediction heads. The independent system achieved the strongest denomination recognition. However, the shared model improves the condition macro-F1 by approximately 3.5 percentage points (pp) and reduces the trainable parameter count by approximately 87%. This improvement is most pronounced for the minority Rare class, suggesting that shared visual features can support the more difficult condition recognition task. Together, the dataset, evaluation protocol, and analysis establish a practical baseline for understanding Bangladeshi banknotes under real-world variations.

Index Terms—Bangladeshi banknotes, denomination recognition, condition classification, multi-task learning, transfer learning, focal loss, assistive technology.

I. INTRODUCTION

Paper currency remains an essential medium for financial transactions, making automatic banknote analysis important for cash-handling systems, vending machines, mobile banking, note-sorting equipment, and assistive technologies for visually impaired users. Unlike laboratory or scanner-based settings, real-world smartphone images often exhibit perspective distortion, illumination changes, shadows, background clutter, occlusions, folds, and device-dependent color variations. Such factors substantially increase the intra-class variation and make reliable recognition considerably more challenging.

Research on Bangladeshi banknote recognition has evolved from handcrafted feature-based methods and conventional neural networks to lightweight deep learning models suitable for mobile deployment [1–5]. Recently, several datasets have been introduced for denomination recognition, banknote detection, and counterfeit analysis [6, 9, 10]. However, each of these datasets is designed to support only one such task in isolation, and none treats the physical condition of banknotes as an explicit learning objective. The condition is instead left as an uncontrolled source of appearance variation.

This study addresses this gap using a real-world dual-label Bangladeshi banknote dataset and evaluates independent and shared learning strategies. Each image contained

both denomination and condition labels, allowing the same visual observation to support two related recognition tasks. The dataset was collected using multiple smartphone cameras across indoor and outdoor environments, daylight and low-light conditions, different backgrounds, and old and new note categories. The final dataset contained 4,260 original images and covered BDT denominations of 10, 20, 50, 100, 200, 500, and 1000.

This study further compared the two practical learning strategies. The independent baseline models were trained separately: ResNet-50 for denomination recognition [20] and MobileNetV3-Large for condition classification [21]. In contrast, the proposed approach adopts a hard parameter-sharing multi-task learning architecture [16], where a shared MobileNetV3-Large backbone is followed by task-specific prediction. Both approaches used ImageNet pre-training [22], class-balanced focal loss [18], AdamW optimization [19], stratified five-fold cross-validation, and data augmentation applied exclusively to the training split. Therefore, this study makes four main contributions. First, this study introduced a self-collected Bangladeshi banknote dataset with joint denomination and condition labels captured under multiple real-world variations. Second, a reproducible preprocessing and stratified cross-validation protocol that preserves the denomination condition distribution across folds while preventing augmented validation samples from entering training is proposed. Third, independent task-specific transfer learning models with a compact shared multi-task learning architecture for joint recognition. Finally, a class-level analysis shows that shared learning improves the more difficult condition recognition task, particularly for the minority Rare class, while maintaining strong denomination recognition performance.

II. LITERATURE REVIEW

A. Banknote Recognition and Assistive Systems

Automatic banknote recognition has evolved from handcrafted feature-based methods to deep learning using RGB cameras and other sensing modalities. The major challenges include illumination changes, orientation variation, physical degradation, and complex backgrounds [11]. Early Bangladeshi systems relied on handcrafted features and conventional neural network classifiers [1]. Recent studies have adopted transfer learning with lightweight CNN architectures such as MobileNet, NASNetMobile, and ResNet for denomination recognition and mobile assistive applications [2–5]. These advances have also facilitated the development of integrated assistive systems that combine banknote recognition

